

SAHAS UPPALAPATI

linkedin.com/in/sahas-uppalapati/

EDUCATION

Ohio State University - Columbus

August 2024 - December 2026

Bachelor's, Computer Science

GPA: 4.0

- Relevant Coursework: Software II, Systems II (Operating Systems), Foundations II (Data structures and Algorithms), Databases, Networking, Multivariate Calculus, Linear Algebra, Statistics for Engineers
- Awards: Dean's List, MakeOHIO Honda Challenge Medalist, Ohio Governors Merit Scholarship, NMSQT Finalist

PROFESSIONAL EXPERIENCE

Amazon

June 2026 - Present

Software Development Engineer Intern

Seattle, WA, USA

- Reduced validation time by 84% by designing parallel execution and concurrency control mechanisms for a Python distributed system using AWS Step Functions, asynchronous workflows, and semaphores across 10K+ weekly customers
- Engineered scalable PySpark aggregation pipelines and data schemas processing 500M+ daily records across 10+ TB of data
- Built a scheduled LangChain LLM agent evaluating recommendations, storing reports in S3 and anomalies in DynamoDB

Nokia

January 2026 - May 2026

Data and Machine Learning Engineer Co-op

Sunnyvale, CA, USA

- Built Python, Kafka and GCP Vertex AI infrastructure supporting CNN training and inference on 2K+ TPS of 5G telemetry
- Automated Kubernetes (GKE) deployment and validation with Docker, Bash, and CI/CD, reducing manual testing by 70%
- Integrated observability with Prometheus, Datadog, Grafana, and Fluent Bit to monitor system health, logs, and performance

Amazon

June 2025 - August 2025

Software Development Engineer Intern

Seattle, WA, USA

- Built a full stack recommendation systems diagnostics tool with React, Java, and AWS reducing debugging time by 50%
- Developed serverless ETL data pipelines using Java REST APIs on AWS Lambda to analyze logs generated by an 8k TPS recommendation system, perform statistical analysis, and validate recommendation quality for a tool adopted by 30+ engineers
- Reduced Athena query costs by 80% through SQL tuning, caching, table partitioning, and Apache Parquet optimization

PROJECTS & OUTSIDE EXPERIENCE

Parallel DAG Workflow Engine

May 2026 - July 2026

- Built a C++20 parallel DAG workflow engine with worker pools and dependency scheduling for graphs up to 1024 nodes
- Benchmarked layered-graph execution, scaling throughput 1.66x over a single-threaded reference at 4 worker threads
- Implemented bounded queues, cancellation, and race-safe shutdown validated with ThreadSanitizer and 67 tests

Distributed Model Serving System

March 2026 - April 2026

- Built Python distributed inference service with ONNX, FastAPI, gRPC, and Protobuf, serving 2 workers at 366 requests/sec
- Reduced inference overhead with dynamic batching, backpressure and bounded queues delivering a p99 latency of 45.56 ms
- Achieved 99%+ service availability in a Docker microservices architecture via health checks, automated retries, and failover

High-Performance Distributed Event Streaming Engine

December 2025 - February 2026

- Built a Linux C++20 UDP event-streaming engine with epoll, nonblocking sockets, a custom binary protocol, and bounded producer-consumer queues for deterministic routing and aggregation across 1,024 concurrent streams and worker threads
- Batched UDP events per datagram to cut per-event overhead, sustaining 1.63M events/sec at 0% loss and 97 μ s p99 latency
- Implemented graceful shutdown, overload handling, and concurrency validation with 135 unit, stress, and sanitizer tests

Infrared Object Detection Pipeline

March 2025 - March 2025

- Fine-tuned YOLOv8n with PyTorch/CUDA on 5,548 infrared images, achieving 85.9% mAP50 on validation data sets
- Built a COCO-to-YOLO image annotation conversion pipeline updating 6,546 person bounding boxes into normalized labels
- Implemented GPU training, evaluation, checkpointing, and batch inference using Ultralytics, measuring 3.1 ms per image

SKILLS

Skills: Python, Java, C++, C, SQL, JavaScript, Bash, AWS, GCP, React, Linux, LangChain, MongoDB, DynamoDB, PostgreSQL, NoSQL, CUDA, TCP, UDP, NumPy, Pandas, Git, Kafka, Apache Spark, Docker, Kubernetes, Jenkins, CI/CD, Agile, Terraform